

Solution Requirements

Date **05 July 2026**

Team ID *********

Project Name **StudyAI – AI Powered Smart Study Companion**

Maximum Marks 3 Marks

Purpose

The purpose of StudyAI is to simplify the learning process by using Artificial Intelligence to automatically convert study materials into useful learning resources such as summaries, flashcards, quizzes, and study plans. The system minimizes manual effort, improves revision efficiency, and enhances students' overall learning experience.

Functional Requirements

FR-1: User can upload study materials

- The system shall allow users to upload PDF, DOCX, and TXT files.
 - Uploaded files are securely stored on the server.
 - Each uploaded file is assigned a unique filename using UUID.
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FR-2: AI Summary Generation

- The system shall extract text from uploaded PDF documents.
 - The extracted text shall be processed using the Groq API.
 - The AI shall generate concise summaries highlighting the key concepts.
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FR-3: Flashcard Generation

- The system shall generate question-answer flashcards from uploaded study material.
 - Flashcards should assist students in active recall and revision.
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FR-4: Quiz Generation

- The system shall automatically create multiple-choice questions (MCQs) from uploaded content.
 - Users should be able to attempt quizzes for self-assessment.
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FR-5: Study Planner

- The system shall generate a personalized study schedule based on user inputs.
 - The planner shall help students manage subjects and available study time efficiently.
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FR-6: Dashboard

- The system shall provide an overview of uploaded materials.
 - Users can access summaries, flashcards, quizzes, and study plans from a centralized dashboard.
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Non-Functional Requirements

Performance

- Generate summaries within a few seconds.
 - Support smooth file uploads with minimal delay.
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Usability

- Simple and intuitive user interface.
 - Easy navigation between modules.
 - Minimal learning curve for new users.
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Reliability

- Ensure successful file uploads.
 - Handle invalid file formats gracefully.
 - Provide appropriate error messages for failures.
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Scalability

- Support increasing numbers of uploaded documents.

- Easily extendable with additional AI-powered learning modules.
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Security

- Validate uploaded file types.
 - Store files securely.
 - Protect API keys using environment variables (.env).
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Hardware Requirements

Client System

- Processor: Intel Core i3 / AMD Ryzen 3 or higher
 - RAM: 4 GB minimum
 - Storage: 500 MB free space
 - Internet Connection
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Server

- Python 3.13+
 - Flask Web Server
 - Internet connectivity for Groq API
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Software Requirements

Frontend

- React.js
 - JavaScript
 - HTML5
 - CSS3
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Backend

- Python
- Flask

- Flask-CORS
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AI

- Groq API
 - Llama Language Model
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PDF Processing

- PyMuPDF (fitz)
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Development Tools

- Visual Studio Code
 - Git
 - GitHub
 - Postman (API Testing)
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Expected Outcome

The proposed StudyAI system will provide students with a centralized AI-powered learning platform capable of reducing study preparation time by automatically generating summaries, flashcards, quizzes, and study plans. This improves learning efficiency, knowledge retention, and overall academic performance.